INCH-POUND
MIL-DTL-24643/62B
27 November 2023
SUPERSEDING
MIL-DTL-24643/62A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +176 DEGREES FAHRENHEIT (+80 DEGREES CELSIUS), 150 VOLTS, TYPE LSPB2SD (INCLUDING VARIATIONS LSPB2SDW, LSPB2SDOS, AND LSPB2SDOSW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable is a single or double shield twisted pair intended for use with PROFIBUS data communication systems. This cable can be manufactured in four variations: LSPB2SD (single shield with drain, non-watertight), LSPB2SDW (single shield with drain, watertight), LSPB2SDOS (double shield with drain, non-watertight), and LSPB2SDOSW (double shield with drain, watertight).

Construction (Watertight and Non-Watertight)

First	-	Copper conductor, solid, uncoated, ASTM B3, 22 AWG, 0.0253 inch nominal. Diameter of conductor may vary from nominal as necessary in order for the completed cable to meet the specified electrical requirements.
Second	-	Polyolefin insulation. Wall thickness as necessary in order for the completed cable to comply with the electrical requirements specified herein. Colored insulation, one red and one green conductor.
Third	-	Two insulated conductors shall be twisted together with a lay as required for the completed cable to comply with the electrical requirements specified herein. Fillers or other means may be used as necessary to form a firm, well-rounded cable that complies with the electrical requirements specified herein.
Fourth	-	An optional binder.
Fifth	-	Copper drain wire, tin-coated, 24 AWG, solid or stranded, ASTM B286.
Sixth	-	Braided shield of round, tin-coated copper, in contact with drain wire. For types LSPB2SD and LSPB2SDW only, AWG and coverage as necessary to comply with shield performance requirements. For types LSPB2SDOS and LSPB2SDOSW, braid shield shall be 34 AWG nominal, 90 percent minimum coverage.
Seventh	-	An optional binder.

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Construction for Variant Types LSPB2SD and LSPB2SDW

Eighth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Construction for Variant Types LSPB2SDOS and LSPB2SDOSW

Eighth - Cross-linked polyolefin jacket (black). Absolute minimum average wall thickness 0.020 inch. Inner jacket surface marking is not applicable.

Ninth - An optional binder.

Tenth - Braided shield of tin-coated copper.

Eleventh - An optional binder.

Twelfth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details.

Military part no. M24643/62	Type and size	Overall diameter (inch)	
		min.	max.
-01UO	LSPB2SD-1	0.330	0.390
-02UO	LSPB2SDW-1	0.345	0.405
-03UD	LSPB2SDOS-1	0.440	0.500
-04UD	LSPB2SDOSW-1	0.450	0.530

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/100 meters at 68 °F [20 °C], max.)	5.64
Conductor resistance unbalance (percent, max.)	5
Jacket flaws	No failure
Conductor and shield continuity	No failure
Voltage withstand	
Conductor to conductor	1.5 kVAC or 2.5 kVDC
Conductor to overall shield	1.5 kVAC or 2.5 kVDC

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Between shields (types LSCPB2SDOS and LSPB2SDOSW only)	1.5 kVAC or 2.5 kVDC
Insulation resistance (megohms/100 meters, max.)	Not applicable

Group A:

Visual and dimensional	No failure
Watertightness (types LSPB2SDW and LSPB2SDOSW only) (see MIL-DTL-24643 for limits of water leakage)	No failure
Mutual capacitance (nominal at 1 kilohertz [kHz]) (nanofarad [nF]/100 meters)	3.0
Input impedance (at 0.2 megahertz [MHz] to 20 MHz) (ohms)	150±15
Swept electrical parameters	See table II

TABLE II. Swept electrical parameters.

Frequency (MHz)	0.2	4.0	16	31.25
Attenuation dB/100 m (max.)	1.0	2.5	5.0	7.2

Group B:

Cold bending (cable)	No failure
Drip (203±1.8 °F [95±1 °C]) (types LSPB2SDW and LSPB2SDOSW only)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	160
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Thermoset proof test (percent, max.)	
Insulation	Not applicable
Jacket (when tested at 392 °F [200 °C])	50

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Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Cable fill (sealant) removability (types LSPB2SDW and LSPB2SDOSW only)	No failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (jacket) (cycles, min.)	125
Shield (conformance to material, construction, and coverage) (Inner shield of types LSPB2SDOS and LSPB2SDOSW only)	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (203±3.6 °F [95±2 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket (overall cable) (also applicable to inner jacket of types LSPB2SDOS and LSPB2SDOSW)	2
Fillers	8
Insulation	18

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Halogen content (percent, max.)	
Jacket (overall cable) (also applicable to inner jacket of types LSPB2SDOS and LSPB2SDOSW)	0.2
Fillers	0.2
Insulation	0.2
Immersion (overall jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket (overall cable) (also applicable to inner jacket of types LSPB2SDOS and LSPB2SDOSW)	25
Fillers	55
Insulation	45
Toxicity index (max.)	
Jacket (overall cable) (also applicable to inner jacket of types LSPB2SDOS and LSPB2SDOSW)	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (overall jacket only)	No failure
Electrical moisture absorption	Not applicable
Stress resistance	Not applicable
Temperature humidity	Not applicable

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All types and sizes	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-078)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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